Blinkit Delivery Delay Root Cause Analysis

Quick Commerce · 10-Minute Delivery · Operations Intelligence

1. Why This Problem Matters

Blinkit's biggest competitive advantage is speed. The **10-minute delivery promise** is a core part of the customer experience and a key reason customers choose quick-commerce platforms.

Recently, this promise has come under pressure. Average delivery times have increased significantly, putting the 10-minute SLA at risk and raising concerns around customer satisfaction, cancellations, retention, and overall trust in the service.

The challenge is that there are several possible explanations for the deterioration, and different teams have different views.

- **Operations:** "We need more riders. This is a supply problem."
- **Marketing:** "Demand has grown too quickly. We should slow down acquisition until operations catch up."
- **Engineering:** "External factors such as seasonal rain may be driving the delays."

Each explanation sounds reasonable, but none should be accepted without looking at the operational data.

That led to the central question:

> **What is actually causing the delivery delays, and where should Operations intervene first?**

2. The Business Context

Blinkit operates in a business where **speed and reliability are part of the product itself**.

A customer choosing quick commerce is not simply buying a product. They are paying for convenience and expecting the order to arrive quickly.

When delivery times increase, the impact can go beyond a few late orders:

This makes the delivery-time problem an **operational and product problem**, not just a logistics issue.

3. The Business Question

The investigation was designed to answer four questions:

1. Is the problem system-wide?

Are all zones experiencing similar deterioration, or is one particular zone responsible for the increase?

2. When does the problem occur?

Is the deterioration spread throughout the day, or concentrated during specific hours?

3. Where is the bottleneck?

Is the delay being created inside the store because of picker capacity, or after dispatch because of rider capacity?

4. Is customer behavior being affected?

Are delivery problems translating into order cancellations?

4. Competing Hypotheses

I started with four possible explanations.

H1 — Rider Capacity

Hypothesis:

Delivery times are increasing because the number of active orders is too high relative to available riders.

What I looked for:

High orders-per-rider during periods of elevated delivery time.

H2 — Store / Picker Capacity

Hypothesis:

Orders are getting delayed inside stores because picker capacity is insufficient relative to order volume.

What I looked for:

High orders-per-picker during periods of elevated delivery time.

H3 — Zone-Level Problem

Hypothesis:

A small number of zones are responsible for most of the delivery degradation.

What I looked for:

Large differences in average delivery time and SLA breach rate across zones.

H4 — External Factors

Hypothesis:

External events such as rain may be contributing to the delivery-time increase.

What I looked for:

A measurable difference in delivery performance during external-event periods.

Data limitation:

Weather/event data was not available in the dataset used for this analysis, so the Rain hypothesis could not be directly validated.

5. My Role

I approached the problem as a **Product Analyst in the Operations Intelligence team**.

My role was not to decide which stakeholder was correct beforehand. Instead, I used the operational data to identify:

- Where delivery performance is deteriorating
- When the deterioration is happening
- Which operational resources are under pressure
- Whether the issue is isolated or system-wide
- Whether customer cancellations are emerging alongside the delivery problem

The goal was to turn a broad operational concern into a **specific, data-backed recommendation**.

6. Data Used

The analysis used three operational datasets:

`blinkit_orders`

Used to analyze:

- Order timing
- Promised ETA
- Actual delivery time
- Delivery status
- Zone-level performance
- Customer cancellations

`blinkit_events`

Used to understand:

- Customer funnel progression
- User activity across different stages
- Progression from home to delivery

`blinkit_store_load`

Used to analyze:

- Active orders
- Available pickers
- Available riders
- Orders per rider
- Orders per picker
- Zone and hour-level operational pressure

7. North Star Metric

10-Minute SLA Adherence Rate

Definition:

> Percentage of orders delivered within 10 minutes.

Why this metric?

The core problem is not simply that delivery times have increased. The bigger issue is that Blinkit's **10-minute customer promise is being missed**.

Therefore, SLA adherence is the clearest metric for measuring whether the operation is delivering the value customers expect.

8. Secondary Metrics

To diagnose the reasons behind SLA deterioration, I tracked four supporting metrics:

Average Delivery Time

Shows the overall speed of the delivery operation.

SLA Breach Rate

Shows how frequently the 10-minute promise is being missed.

Orders per Rider

Measures road-side capacity pressure.

Formula:

> Active Orders ÷ Available Riders

Orders per Picker

Measures store-level capacity pressure.

Formula:

> Active Orders ÷ Available Pickers

9. Analytical Approach

I used SQL to break the problem down from the overall business level into specific operational patterns.

Step 1 — Measure the scale of the problem

Calculate average delivery time and SLA breach rate.

Step 2 — Compare zones

Determine whether one zone is responsible for the deterioration.

Step 3 — Analyze by hour

Identify when delivery performance deteriorates most.

Step 4 — Compare demand with capacity

Calculate orders per rider and orders per picker to identify operational pressure.

Step 5 — Examine customer behavior

Analyze cancellation rates and funnel progression.

Step 6 — Evaluate competing hypotheses

Compare the evidence against the proposed explanations and identify which hypotheses are supported by the available data.

10. What I Wanted to Find

The objective was not simply to report that deliveries were late.

I wanted to answer a more useful operational question:

> ✨ **What is happening, where is it happening, when is it happening, and what can Operations change to prevent it?**

The final recommendation would therefore focus on the **specific operational constraint identified by the data**, rather than applying a blanket solution across the entire network.

Project Goal

> ✨ **Identify the root cause of the delivery SLA breach and recommend a targeted operational intervention that improves delivery reliability without unnecessarily increasing operational costs.**